

Lecture 8: Multi-Armed Bandit

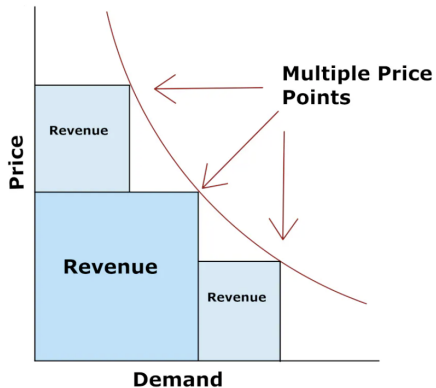
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Multi-Armed Bandit



- States: fictive initial and final states
- Actions: to choose an arm
- Reward: points

Example: Dynamic Pricing



- States: fictive initial and final states
- Actions: to set the price
- Reward: revenue

Multi-Armed Bandit

One-State Decision Process $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{P}_0, \mathcal{R}, \gamma \rangle = \langle \mathcal{A}, \mathcal{R} \rangle$

- $\mathcal{S} = \{s_0\}$ is a fictive one-state space
- $\mathcal{A} = \{a_i: i \in 1, m\}$ is a finite action space
- $\mathcal{P}$ is a fictive transition probability function $\mathcal{P}(s_0|s_0, a) = 1$
- $\mathcal{P}_0$ is a fictive initial state probability function $\mathcal{P}(s_0) = 1$
- $\mathcal{R}$ is an unknown reward **probability function**

$$\mathcal{R}(r|a) = \mathbb{P}[R_t = r | A_t = a]$$

- $\gamma = 1$ is a fictive discount coefficient

The problem is to maximize reward sum:

$$\mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} R_t \right] \longrightarrow \max_{\pi}$$

Problem Statement

Best Reward and Optimal Action

$$r_* = \max_{a \in \mathcal{A}} \mathbb{E}[\mathcal{R}(\cdot|a)], \quad a_* = \operatorname{argmax}_{a \in \mathcal{A}} \mathbb{E}[\mathcal{R}(\cdot|a)]$$

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If $r_* > 0$, then $\forall \pi : \pi(a_*) > 0 \Rightarrow$

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Regret Formalism

$$\mathbb{E}_\pi[\rho_\infty] \longrightarrow \max_\pi, \quad \rho_\infty = \sum_{t=0}^{\infty} (r_* - R_t)$$

then $\exists \pi_*$, but **there is no algorithm to make $\mathbb{E}_\pi[\rho_\infty]$ finite**. Therefore, we investigate the behavior of

$$\mathbb{E}_\pi[\rho_T], \quad \rho_T = \sum_{t=0}^T (r_* - R_t)$$

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- Set π and $D_0 = \emptyset$
- Agent acts $A_0 \sim \pi(\cdot|D_0)$
- gets the reward $R_0 \sim \mathcal{R}(\cdot|A_0)$ and update $D_1 = (A_0, R_0) \cup D_0$

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- Regret sequence $(\rho_0, \rho_1, \rho_2, \dots)$

Action-Value Function

One-Step Q-Function

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Q-Function Approximation

$$q^{(1)}(a) \approx Q_T(a) := \frac{H_T(a)}{N_T(a)},$$

where

$$H_T(a) = \sum_{R \in \mathfrak{R}_T(a)} R, \quad N_T(a) = |\mathfrak{R}_T(a)|, \quad \mathfrak{R}_T(a) = \{R : (a, R) \in D_T\}$$

Naive Policies

Choice of constant action

$$\pi(a) = \begin{cases} 1, & \text{if } a = \hat{a} \\ 0, & \text{otherwise} \end{cases}$$

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Random choice for each episode

$$\pi(a) = 1/m$$

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ε -Greedy Policies

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$$\pi(a|Q_T) = \begin{cases} 1 - \varepsilon + \varepsilon/m, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} Q_T(a') \\ \varepsilon/m, & \text{otherwise} \end{cases}$$

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$$\mathbb{E}_\pi[\rho_T] \leq c_\omega \log T$$

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$$\pi(a|Q_T) = \text{SoftMax}(Q_T/\theta), \quad \theta > 0$$

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Upper Confidence Bound

UCB

$$\pi(a|Q_T, U_T) = \begin{cases} 1, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} (Q_T(a') + U_T(a')) \\ 0, & \text{otherwise} \end{cases}$$

where $Q_T(a) + U_T(a) \geq q(a)$ with probability e.g. $T^{-2\kappa}$ (usually $\kappa = 2$). From the Hoeffding's inequality, we derive

$$U_T(a) = \sqrt{\kappa \frac{\log(T)}{N_T(a)}},$$

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Thompson Sampling

Binary Reward Case

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$$\pi(a|\alpha_T, \beta_T) = \begin{cases} 1, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} \xi(a'), \\ 0, & \text{otherwise} \end{cases}$$

where

$$\xi(a) \sim B(\alpha_T(a), \beta_T(a)), \quad \alpha_T(a) = H_T(a), \quad \beta_T(a) = N_T(a) - H_T(a)$$

$$H_T(a) = \sum_{R \in \mathfrak{R}_T(a)} R, \quad N_T(a) = |\mathfrak{R}_T(a)|, \quad \mathfrak{R}_T(a) = \{R : (a, R) \in D_T\}$$

QUESTIONS?